

LONDON INTL, ON, Canada

WMO: 716230

Lat: 43.0269N

Lon: 81.1486W

Elev: 278

StdP: 98.03

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.0	-16.1	-23.1	0.5	-18.4	-19.9	0.7	-15.5	12.7	-5.1	11.7	-5.1	3.2	0	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	30.3	22.3	28.8	21.6	27.3	20.7	23.8	28.1	22.8	27.0	22.0	25.8	4.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.5	17.8	26.3	21.5	16.8	25.2	20.6	15.8	24.4	72.9	28.3	69.0	26.9	65.7	25.7	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.0	9.5	8.4	-22.8	32.8	3.4	1.9	-25.3	34.1	-27.3	35.2	-29.2	36.3	-31.7	37.7		
			-23.1	26.0	3.4	1.2	-25.6	26.9	-27.5	27.6	-29.4	28.3	-31.9	29.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	-5.4	-4.6	0.1	6.9	13.6	18.9	21.0	20.1	16.6	10.2	3.7	-1.8
	DBStd	10.53	6.34	5.95	6.03	4.84	4.62	3.69	3.00	2.82	3.84	4.50	4.75	5.09
	HDD10.0	1948	478	408	311	116	17	0	0	0	2	54	194	367
	HDD18.3	3899	736	642	564	342	161	36	8	14	77	256	438	624
	CDD10.0	1348	0	0	6	24	128	268	342	314	199	59	6	1
	CDD18.3	258	0	0	0	1	13	54	92	70	26	2	0	0
	CDH23.3	1979	0	0	1	7	124	431	724	481	197	14	0	0
	CDH26.7	446	0	0	0	0	21	108	181	93	41	2	0	0
Wind	WSAvg	3.9	4.8	4.7	4.5	4.5	3.9	3.3	2.9	2.8	3.1	3.7	4.2	4.4
Precipitation	PrecAvg	1015	90	67	76	78	89	90	90	76	91	86	90	90
	PrecMax	1282	132	130	154	146	182	194	176	174	291	167	159	154
	PrecMin	737	52	19	31	30	18	42	19	12	17	22	27	42
	PrecStd	151	23	27	32	29	35	35	36	33	56	36	34	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	12.5	20.8	24.3	29.1	31.3	32.3	31.5	30.4	25.1	17.7	13.0
		MCWB	10.8	10.1	16.1	16.4	20.1	22.1	24.0	22.7	21.9	19.1	14.2	11.3
	2%	DB	8.4	8.0	15.3	20.8	26.7	29.6	30.4	29.0	27.6	22.1	15.1	9.7
		MCWB	7.7	6.3	10.9	14.1	19.3	21.8	22.6	22.3	20.7	17.3	12.3	8.4
	5%	DB	5.1	5.1	12.0	18.0	24.3	27.7	28.7	27.4	25.6	19.7	13.3	7.1
		MCWB	4.0	3.3	9.2	12.2	17.7	20.7	21.7	21.1	19.5	15.9	10.9	5.7
	10%	DB	2.7	2.9	8.8	15.3	21.8	25.8	27.1	26.1	23.6	17.5	11.2	4.6
		MCWB	1.7	1.5	6.0	10.2	15.8	19.5	20.7	20.4	18.6	14.4	8.9	3.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.8	10.9	16.3	17.6	22.0	24.1	25.6	25.0	23.9	20.5	15.1	11.4
		MCDB	11.6	12.2	20.9	22.3	27.4	29.3	30.5	28.1	28.1	23.8	16.8	12.4
	2%	WB	7.9	6.7	12.0	15.3	20.2	22.8	24.0	23.4	22.1	18.3	13.0	8.7
		MCDB	8.3	7.9	14.4	19.2	25.2	27.8	28.2	27.2	25.8	20.9	14.7	9.4
	5%	WB	4.2	3.5	9.2	13.3	18.7	21.7	22.9	22.4	20.8	16.4	11.1	5.8
		MCDB	5.0	4.9	11.9	16.9	22.8	26.1	26.9	25.9	23.9	19.0	12.9	6.8
	10%	WB	1.7	1.6	6.1	10.9	17.0	20.6	22.0	21.4	19.5	14.8	9.1	3.5
		MCDB	2.6	2.8	8.5	14.3	20.8	24.6	25.6	24.7	22.5	17.1	11.0	4.5
Mean Daily Temperature Range	5% DB	MDBR	7.0	7.7	8.3	10.1	10.6	10.2	10.3	10.2	10.6	8.9	7.2	5.9
		MCDBR	8.6	9.9	12.5	14.4	13.3	11.8	11.6	11.4	12.8	11.7	10.3	8.2
		MCWBR	8.4	8.7	9.4	9.5	7.8	6.2	5.9	6.1	7.0	7.5	8.2	7.5
	5% WB	MCDBR	8.3	9.6	11.9	12.7	11.8	10.3	9.8	9.6	10.5	10.4	9.5	8.0
		MCWBR	8.5	8.9	9.7	9.6	7.8	6.3	5.8	5.9	6.8	7.4	8.2	7.7
Clear-Sky Solar Irradiance	taub	0.293	0.327	0.332	0.369	0.395	0.426	0.415	0.399	0.373	0.357	0.315	0.289	
	taud	2.391	2.262	2.384	2.316	2.284	2.223	2.278	2.304	2.371	2.400	2.482	2.457	
	Ebn at Noon	844	861	910	897	878	849	853	856	854	821	814	818	
	Edh at Noon	78	105	106	122	130	139	130	123	106	91	72	67	
All-Sky Solar Radiation	RadAvg	1.44	2.12	3.50	4.54	5.50	6.16	6.09	5.33	4.23	2.58	1.56	1.02	
	RadStd	0.15	0.30	0.41	0.45	0.49	0.38	0.39	0.27	0.38	0.23	0.23	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page